

Program : Diploma in Cyber Forensics And Information security	
Course Code : 5283B	Course Title: IoT Security
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4T:0 P:0)	Periods per semester: 60

Course Objectives:

- Understand the basics of IoT, sensors, actuators and IoT boards in real time environment.
- Learn arduino programming
- Familiarize analog and digital communication
- Understand the Security requirements in IoT

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic Knowledge in information systems.		Introduction to IT Systems Lab	I
Basic knowledge in programming.		Problem Solving and Programming	II

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the basics of IoT, sensors, actuators and IoT boards in real time environment	15	Understanding
CO2	Understand the syntax of control statements, operators , I/O functions for problem solving	14	Understanding
CO3	Understand analog and digital communications with arduino	15	Understanding
CO4	Understand the Security requirements in IoT	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2		2				
CO2	2		2				
CO3	-					1	
CO4	1	3	3			1	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the basics of IoT, sensors, actuators and IoT boards in real time environment		
M1.01	Introduction: Introduction, Evolution, IoT versus M2M, CPS, WoT;	3	Understanding
M1.02	Enabled technologies, Networking components, Challenges and applications	3	Understanding
M1.03	Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations	3	Understanding
M1.04	Actuators: Definition and Types	3	Understanding
M1.05	Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard	3	Understanding
Contents: Introduction: Introduction to IoT, Evolution of IoT-IoT versus M2M, IoT versus CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard			
CO2	Understand the syntax of control statements, operators , I/O functions for problem solving		
M2.01	Introduction to arduino	2	Understanding
M2.02	Program elements	2	Understanding
M2.03	statements	2	Understanding

M2.04	Connecting an external led	3	Understanding
M2.05	Working with buttons	3	Understanding
M2.06	Interrupts	2	Understanding
	Series Test – I	1	
Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements -Structure, Variables and constants, Data types, Operators, Control statements , Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts			
CO3	Understand analog and digital communications with arduino		
M3.01	Introduction-Serial communication	2	Understanding
M3.02	Introduction Parallel communication	2	Understanding
M3.03	Interfacing LCD character display	2	Understanding
M3.04	UART Communication	3	Understanding
M3.05	I2C Communication	3	Understanding
M3.06	SPI Communication	3	Understanding
Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication : UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication : I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication : SPI protocol, SD card interfacing, Ethernet module interfacing			
CO4	Understand the Security requirements in IoT		
M4.01	Security Requirements in IoT Architecture	2	Understanding
M4.02	Security Architecture in the Internet of Things	3	Understanding
M4.03	Security Requirements in IoT	2	Understanding
M4.04	Insufficient Authentication / Authorization	2	Understanding
M4.05	Vulnerabilities – Secrecy and Secret-Key Capacity	2	Understanding
M4.06	Transport Encryption	3	Understanding

	Series Test – II	1	
Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities – Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption – Attack & Fault trees			

Text / Reference

T/R	Book Title/Author
T1	Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New Delhi: University Cambridge Press, 2021. (Chapter 4)
T2	Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt Publishing Ltd., 2015. (Chapter 2-6)
R1	Fei HU, “Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations”, CRC Press, 2016
R2	Russell, Brian and Drew Van Duren, “Practical Internet of Things Security”, Packt Publishing, 2016.

Online Resources

Sl.No	Website Link
1	https://www.arduino.cc/en/IoT/HomePage
2	https://learn.adafruit.com/category/internet-of-things-iot